

Software Requirement Specification

Procurement Management System with Odoo 17, XML & OWL

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1. Introduction

1.1 Purpose

This document defines the software requirements for the **Procurement Management System** for the **Youth Skill Development Training Program system**. The system is designed to facilitate supplier registration, RFP (Request for Purchase) management, and quotation evaluation, ensuring streamlined workflows for suppliers and reviewers.

1.2 Scope

The system will allow:

1. Suppliers to register, submit documents, and bid on RFPs.
2. Reviewers to create RFPs, evaluate supplier applications, and review quotations.
3. Approvers to finalize supplier selection and RFP approvals.
4. Automated reports and dashboards to provide insights for decision-making.

1.3 Intended Stakeholder

1. **Suppliers:** Register and participate in RFPs.
2. **Reviewers:** Manage RFPs and evaluate supplier applications.
3. **Approvers:** Make final decisions on supplier approval and RFP selection.
4. **Administrators:** Oversee system management and security.

1.4 References

Reference	Location
Odoo 17	https://www.odoo.com/documentation/17.0/developer.html
Odoo 17 Coding Guidelines	https://www.odoo.com/documentation/17.0/contributing/development/coding_guidelines.html

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
RFP	Request for Purchase
RFQ	Request for Quotation
OWL	Odoo Web Library

2. Overall Description

2.1 Overview

This system will digitize and streamline supplier onboarding, RFP publication, and quotation management. The key objectives include:

1. Efficient supplier registration with OTP verification and document submission.
2. Structured RFP lifecycle from creation to approval.
3. Quotation evaluation and scoring to recommend suppliers.
4. Automated reporting and dashboard insights for stakeholders.

2.2 Technical Platform

1. Backend: Odoo 17 (Python, PostgreSQL)
2. Frontend: OWL (Odoo Web Library), XML Views
3. Deployment: Docker with Nginx reverse proxy
4. Database: PostgreSQL
5. Operating System: Linux (Ubuntu)

3. Functional Requirements

3.1 Overview

Serial No	Main Features	Description
1	Email Verification	OTP-based supplier account creation
2	Supplier Registration	Multi-step form submission with document upload
3	Two-Step Verification	Review and approval process for supplier onboarding
4	RFP Management	Creation, publication, and quotation submission
5	Reporting	Generate supplier-wise RFP reports
6	Data Visualization	Graph and pivot views for RFP insights
7	Dashboard	Interactive OWL-based data analytics

4. User Interface

UI No.	UI Name	Description
1	Supplier Portal	Allows suppliers to register and submit documents
2	Reviewer Panel	Enables reviewers to evaluate applications and manage RFPs
3	Approver Panel	Enables approvers to review and finalize supplier selection
4	Dashboard	Provides key insights through data visualization

5. Non-Functional Requirements

5.1 Performance Requirements

1. OTP verification should be completed within the time limit.
2. System should efficiently handle RFP listing and searches.

5.2 Security Requirements

1. Role-based access control should be enforced.
2. Secure handling of supplier financial and personal data.
3. Blacklist verification and email validation must be implemented.

6. Design Constraints

1. Must follow Odoo 17's module structure.
2. Use OWL for frontend development.
3. Must be Docker containerized.

7. Software Quality Attributes

Requirement ID	Description	Acceptability Criteria
QA-01	Performance	System should respond efficiently for key operations
QA-02	Security	Data encryption and access controls must be implemented

8. Other Requirements

1. Reports should be exportable in Excel and PDF formats.
2. Notifications should be triggered via email and dashboard alerts.
3. Testing should cover unit, integration, and security tests.